Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AI Vibe Coding for Excel VBA

Course Code: C357

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	28 September 2026	First release of the AI Vibe Coding for Excel VBA courseware (C357): five topics and 31 hands-on labs built on the NorthWind Sales Commission Tracker, scheduled 9:30am – 5:30pm (7.5 instructional hours per day, 15 hours in total).	Dr. Alfred Ang

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

TABLE OF CONTENTS

Course Information **4**
Learning Outcomes **4**
Course Schedule **4**
Day 1 — AI-assisted VBA foundations — prompt, generate, inspect and build reusable procedures 4
Day 2 — AI-assisted Excel automation — objects, testing, debugging and user-defined forms 5
Lab Reference (aligned to learning outcomes) **6**
Training Resources and Lab Materials **7**

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Course Information

Course Title	AI Vibe Coding for Excel VBA
Course Code	C357
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 7.5 instructional hours per day (15 hours in total)
Daily Timing	9:30 am – 5:30 pm on both days (30-minute lunch break; refreshments are taken during activity time)
Mode	Instructor-led, hands-on AI-assisted Excel VBA labs on a Windows or Mac PC with Microsoft Excel
Pre-requisites	Working knowledge of Microsoft Excel (formulas, worksheets). No prior programming experience required.
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Design a simple Excel interface using VBA, macros and an AI-assisted coding workflow
- LO2: Prompt AI to generate VBA, then review and adapt the code using appropriate programming methods and tools
- LO3: Create and validate AI-generated functions and procedures that meet design objectives
- LO4: Manage Excel object functionality and interoperability while testing and debugging AI-generated VBA
- LO5: Produce user-defined forms, documentation and auditable AI prompt-to-code evidence

Course Schedule

Day 1 — AI-assisted VBA foundations — prompt, generate, inspect and build reusable procedures

Time	Duration	Topic / Activity
9:30am–10:00am	30 min	Welcome, trainer and learner introductions, ground rules, learning outcomes and course outline
10:00am–11:00am	60 min	**AI vibe-coding playbook plus Topic 1 — frame the task, prompt for VBA, inspect generated code, use the**

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

		Visual Basic Editor, modules, procedures, boxes and the macro recorder
11:00am–12:45pm	105 min	Hands-on: Lab 1: Set Up the Developer Ribbon and Save a Macro-Enabled Workbook; Lab 2: Write Your First Macro in a Module; Lab 3: Message Boxes and Button Choices; Lab 4: Capture User Input with an Input Box
12:45pm–1:15pm	30 min	Lunch break
1:15pm–2:15pm	60 min	Hands-on: Lab 5: Record a Macro to Calculate Sales Tax; Lab 6: Run a Macro from a Button and a Shape
2:15pm–3:15pm	60 min	**Topic 2 — prompt AI with explicit variables, types, business rules and tests; review generated arrays, operators, decisions and loops (concepts + demo)**
3:15pm–5:15pm	120 min	Hands-on: Lab 7: Declare Variables and Constants with the Right Data Type; Lab 8: Build Expressions with VBA Operators; Lab 9: Branch with If…Then…ElseIf…Else; Lab 10: Simplify Banded Logic with Select Case; Lab 11: Store a Column of Figures in an Array; Lab 12: Process Every Row with For…Next and For Each; Lab 13: Find the First Empty Row with Do…While and Do…Until
5:15pm–5:30pm	15 min	Day 1 recap and Q&A

Day total: 7.5 instructional hours (9:30am – 5:30pm) — 450 minutes of teaching plus a single 30-minute lunch break.

Day 2 — AI-assisted Excel automation — objects, testing, debugging and user-defined forms

Time	Duration	Topic / Activity
9:30am–9:50am	20 min	Day 1 recap
9:50am–10:55am	65 min	**Topic 3 — specify and validate AI-generated functions and Subs: contracts, arguments, return values, scope, worksheet functions and charts**
10:55am–1:05pm	130 min	Hands-on: Lab 14: Write a Reusable Function that Returns a Value; Lab 15: Compare a Function with a Sub Procedure; Lab 16: Control Visibility with Public and Private

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

		Scope; Lab 17: Reuse Excel's Built-in Worksheet Functions; Lab 18: Generate a Commission Chart from Code
1:05pm–1:35pm	30 min	Lunch break
1:35pm–2:20pm	45 min	**Topic 4 — review AI-generated object references, error paths and side effects; test and debug Excel objects (concepts + demo)**
2:20pm–3:45pm	85 min	Hands-on: Lab 19: Read and Write Cells with the Range Object; Lab 20: Copy Data Between Worksheets; Lab 21: Automate Workbooks and Worksheets; Lab 22: Identify Syntax, Runtime and Logical Errors; Lab 23: Handle Runtime Errors with On Error GoTo; Lab 24: Debug with Breakpoints, Step Into and the Watch Window; Lab 25: Find a Value and Handle the Not-Found Case
3:45pm–4:05pm	20 min	**Topic 5 — generate, review and document event-driven UserForms, controls and validation with prompt-to-code evidence**
4:05pm–5:10pm	65 min	Hands-on: Lab 26: Respond to Workbook Events; Lab 27: Respond to Worksheet Events; Lab 28: Design a UserForm for Sales Entry; Lab 29: Write the Form's Event Procedures; Lab 30: Validate Form Input and Document the Solution; Lab 31: Capstone — Complete the NorthWind Commission Tracker
5:10pm–5:30pm	20 min	Course recap, capstone review and course feedback

Day total: 7.5 instructional hours (9:30am – 5:30pm) — 450 minutes of teaching plus a single 30-minute lunch break.

Lab Reference (aligned to learning outcomes)

Topic	Learning outcome	Labs
Topic 01: Get Started on VBA	LO1	Lab 1, Lab 2, Lab 3, Lab 4, Lab 5, Lab 6
Topic 02: VBA Programming	LO2	Lab 7, Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13
Topic 03: Function and Sub Procedure	LO3	Lab 14, Lab 15, Lab 16, Lab 17, Lab 18
Topic 04: Managing Excel	LO4	Lab 19, Lab 20, Lab 21, Lab

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Objects		22, Lab 23, Lab 24, Lab 25
Topic 05: Create User Defined Forms	LO5	Lab 26, Lab 27, Lab 28, Lab 29, Lab 30, Lab 31

Training Resources and Lab Materials

The following materials are issued to every learner through the LMS before the course begins:

- Course slide deck (AI Vibe Coding for Excel VBA v1.0) — trainer PPT and learner PDF.
- Learner Guide — AI prompt-to-code workflow, full step-by-step instructions, review checks and worked VBA solutions for all 31 labs.
- This Lesson Plan.
- The labs folder — one sub-folder per lab (31 in total).

Each lab folder contains the mock Excel files for that lab:

File	Purpose
lab-NN-<title>.md	The lab instructions — AI prompt, review checklist, objective, steps, worked code and verification evidence.
lab-NN-start.xlsm	The mock workbook as it stands BEFORE the lab, with a safe embedded lab-launch macro.
lab-NN-end.xlsm	The completed workbook for this lab — the sheets and results as they should look, plus a runnable lab-specific VBA checkpoint embedded for checking work and catching up.
vba/	The worked solution as importable VBA source — .bas standard modules, .cls workbook and worksheet event code, and the .frm UserForm. Learners type the code themselves; importing is the catch-up route.
../data/*.csv	The shared mock datasets — NorthWind sales, tax, targets, shipping and loan figures.

- All 31 labs build ONE workbook (the NorthWind Sales Commission Tracker), so each lab's start workbook is identical to the previous lab's end workbook. A learner who falls behind, joins late or loses a file can open the start workbook of the current lab and continue with the class immediately.
- Trainers should demonstrate from the start workbook of each lab so the class sees the same state on screen as on their own machines.
- Every workbook carries a real embedded VBA project. The start file contains a safe lab-launch macro; the end file contains a runnable lab-specific checkpoint. Complete source in vba/, including event and UserForm code, remains the inspection and import route. Learners must still explain, review and test their own generated code.
- Remind learners at every save point that work must be saved as .xlsm; saving as .xlsx silently discards all macros, forms and button assignments.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg